

ITEC202 — Data Management and Visualisation

Assessment Task 3 Part B: Data Visualisation Project

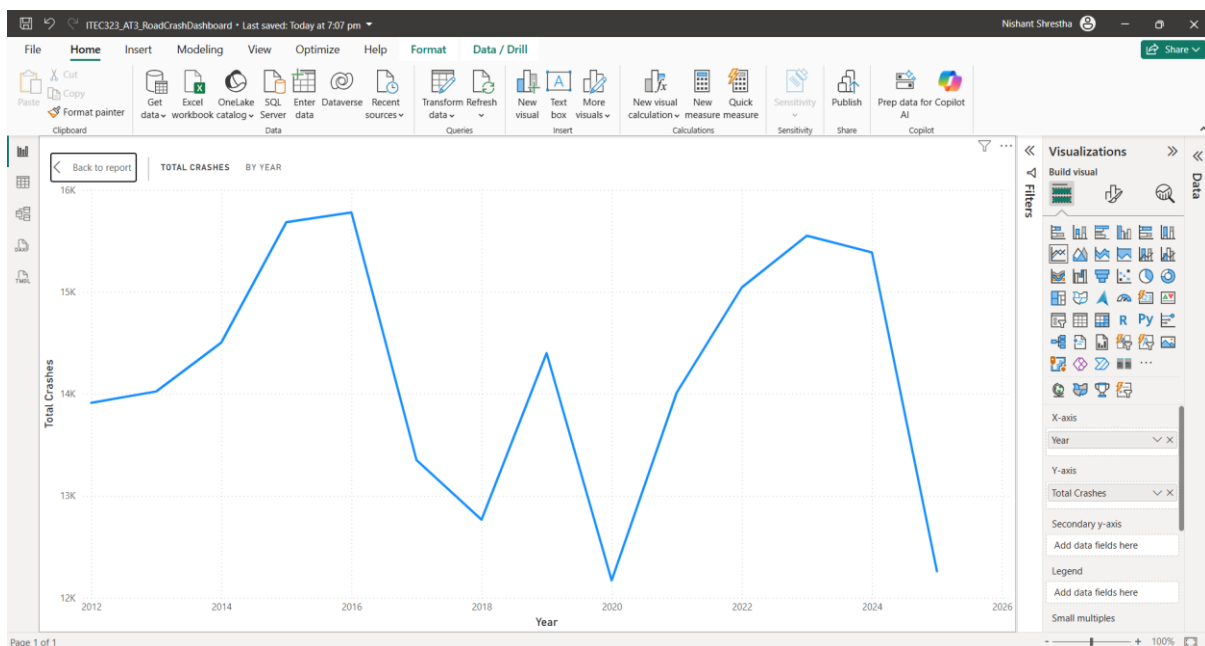
Victorian Road Crash Data Analysis

Field	Details
Unit Code	ITEC202
Unit Name	Data Management and Visualisation
Assessment	Task 3 Part B — Data Visualisation Project
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Visualisations

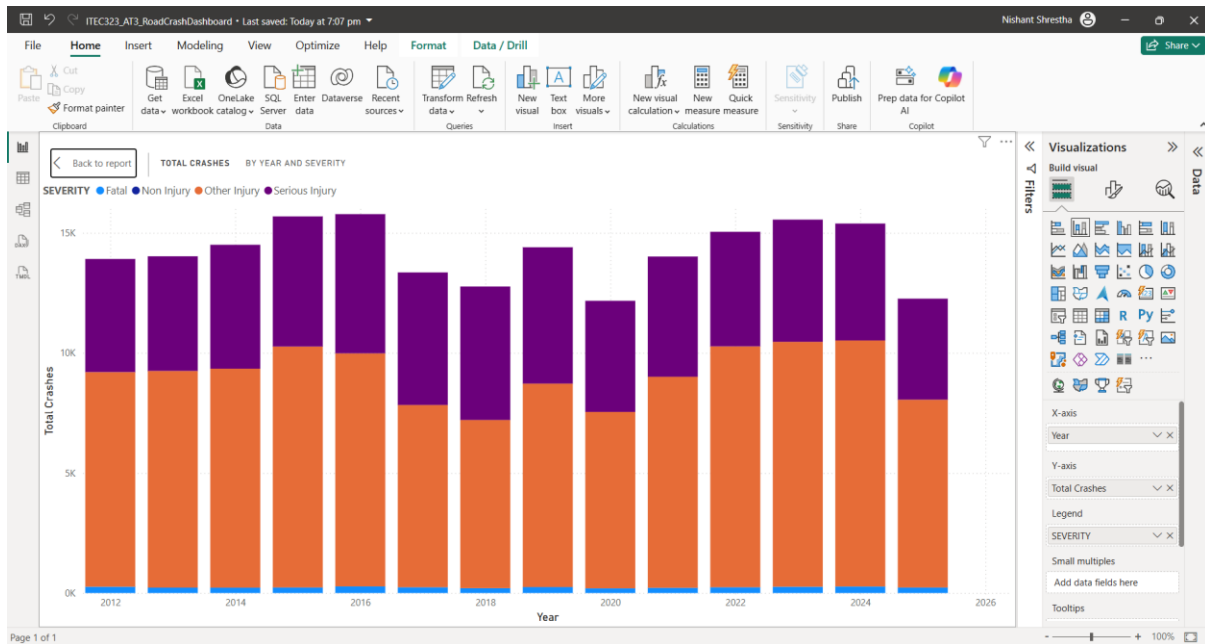
The following four visualisations were created in Microsoft Power BI Desktop using the Victorian Road Crash dataset. Each chart uses data from two or more of the three required tables — Accident, Vehicle, and Person — connected through the star schema data model. The visuals were selected to provide meaningful insights into road crash trends, severity, environmental conditions, and road user risk, in line with the goal of supporting evidence-based road safety policy decisions.

Visual 1 — Annual Crash Trend (Line Chart)



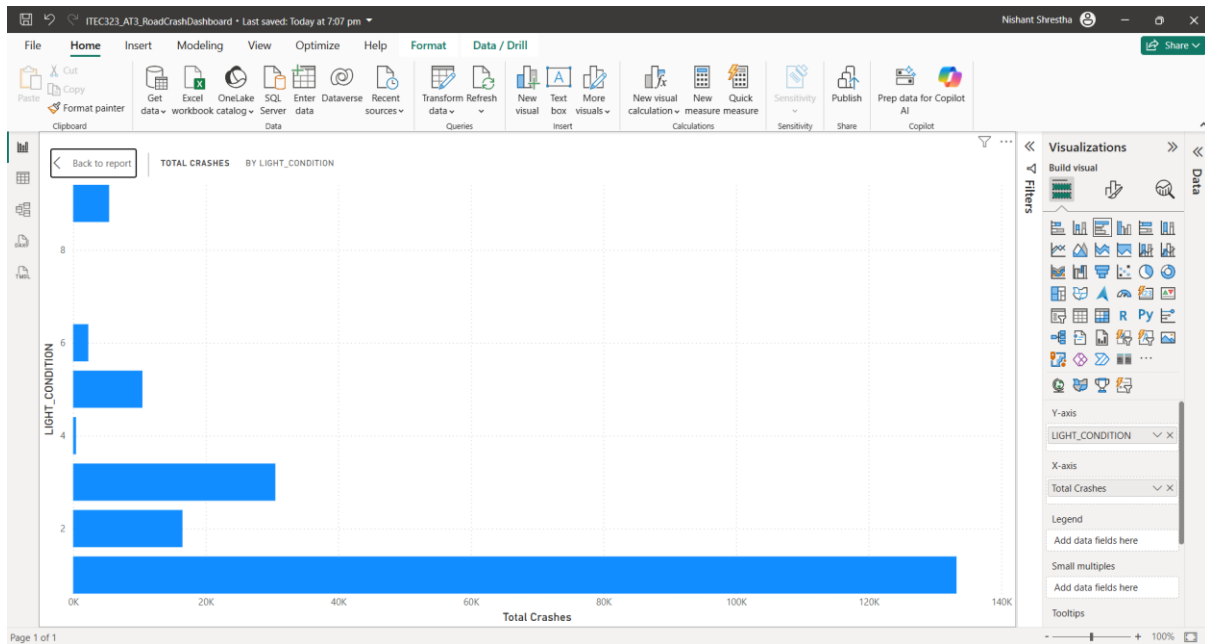
Description: This line chart displays the total number of road crashes in Victoria for each year covered by the dataset. It allows the viewer to identify long-term trends in crash frequency, spot the impact of road safety programs or external events on crash numbers, and evaluate whether crash volumes are rising or falling over time. Understanding these patterns is essential for government agencies when planning road safety investment and policy responses.

Visual 2 — Crash Severity by Year (Stacked Column Chart)



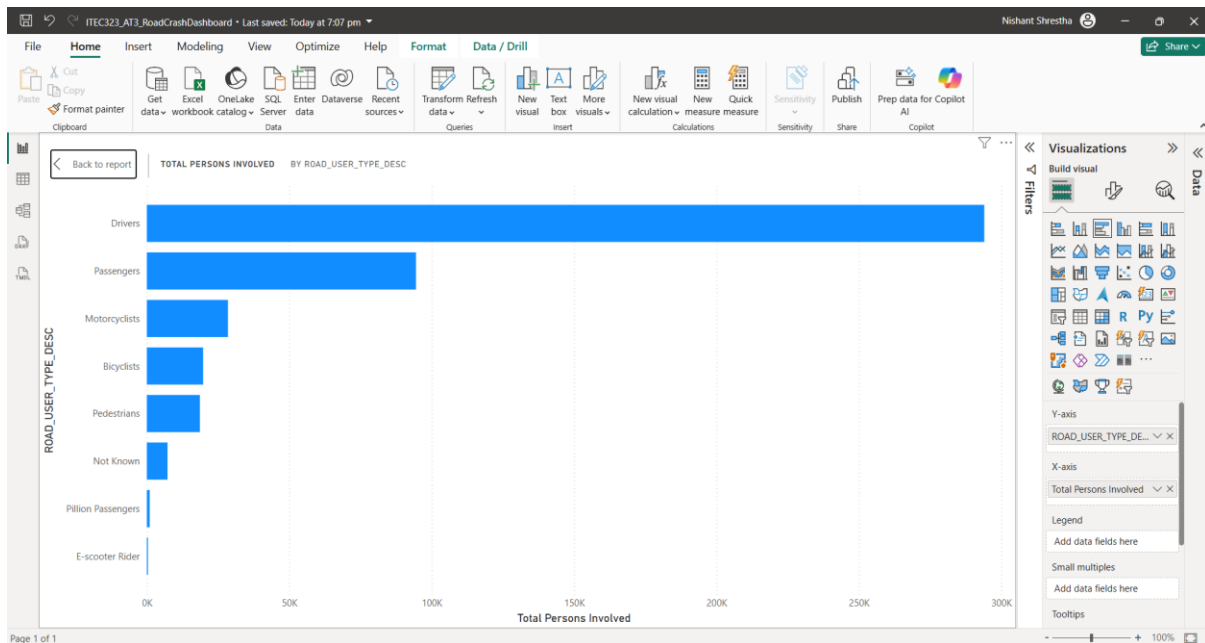
Description: This stacked column chart presents the annual breakdown of crash outcomes by severity level, including Fatal, Serious Injury, Other Injury, and Non Injury. It shows both the total crash volume and the composition of severity categories in each year. This is important because it allows analysts to see whether the proportion of fatal and serious injury crashes is changing over time, which is central to monitoring progress under Victoria's Towards Zero road safety strategy.

Visual 3 — Crashes by Light Condition (Bar Chart)



Description: This horizontal bar chart compares the total number of crashes recorded under different lighting conditions, such as daylight, dusk or dawn, and darkness. It is useful because it reveals whether poor visibility is a significant contributing factor to crash frequency, which in turn informs decisions about street lighting upgrades, road marking improvements, and the placement of speed management infrastructure on high-risk corridors across Victoria.

Visual 4 — Persons Involved by Road User Type (Bar Chart)



Description: This horizontal bar chart shows the total number of persons involved in road crashes broken down by road user category, including Drivers, Passengers, Motorcyclists, Bicyclists, and Pedestrians. This visual draws on data from both the Person table and the Accident table through the established relationship. It is important because it reveals which road user groups face the highest crash exposure, directly informing targeted safety campaigns and active transport infrastructure investment decisions.

Data Model Screenshot

The data model was built using a star schema structure, which is the standard approach for analytical reporting in Power BI. The Accident table serves as the central fact table because it stores one unique row per crash event, identified by the ACCIDENT_NO field, and contains all the key measurable facts including severity, speed zone, time, and date. The Vehicle and Person tables are connected to Accident as dimension tables through one-to-many relationships, using ACCIDENT_NO as the shared linking key between all three tables.

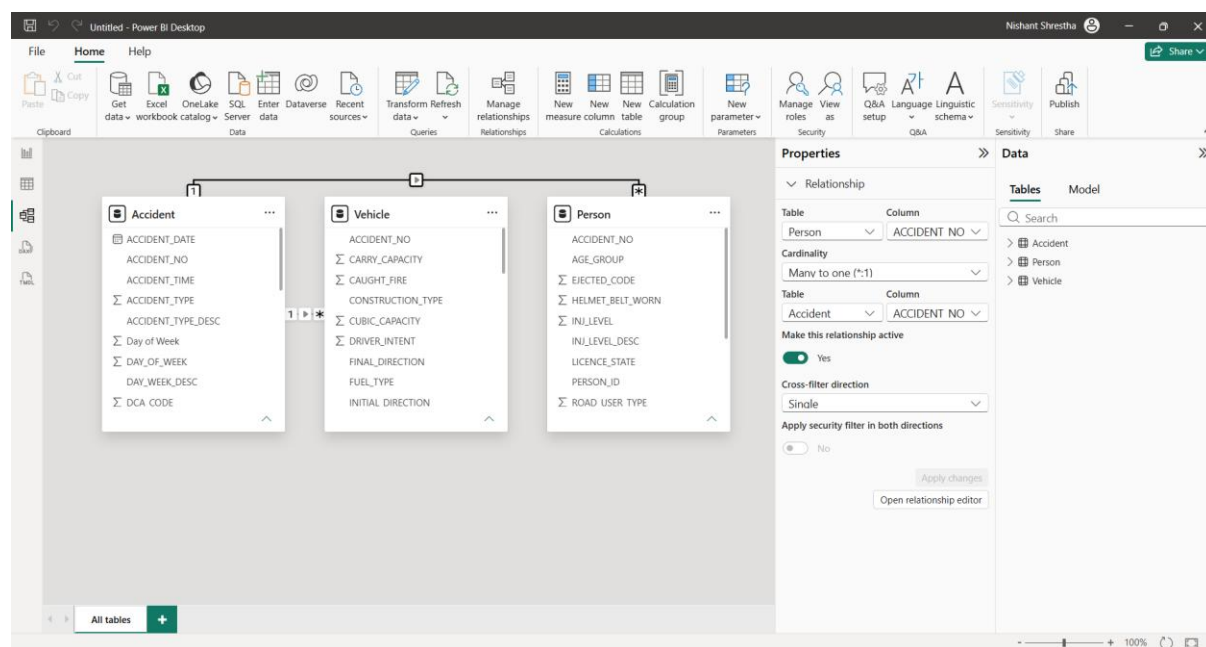


Figure 5: Star schema data model showing the Accident table as the central fact table, connected to Vehicle and Person dimension tables via ACCIDENT_NO. Both relationships use One-to-Many cardinality with Single filter direction.

Relationship	Primary Key	Foreign Key	Cardinality	Filter Direction
Accident → Vehicle	Accident: ACCIDENT_NO	Vehicle: ACCIDENT_NO	One-to-Many (1:*)	Single
Accident → Person	Accident: ACCIDENT_NO	Person: ACCIDENT_NO	One-to-Many (1:*)	Single

Single filter direction was applied to both relationships because all analytical queries in this dashboard originate from the Accident table and flow outward to Vehicle and Person. This follows Microsoft Power BI best practice for star schema design, prevents ambiguous cross-filtering between tables, and improves the performance and reliability of DAX measure calculations throughout the model.

Critical Reflection

Building this dashboard gave me a much clearer appreciation of how data visualisation can convert a large and complex dataset into something that is genuinely useful for decision making. The Victorian Road crash dataset contains close to 200,000 records covering more than a decade of crash history across the state. On its own, that volume of data is impossible to interpret or act on. By using Power BI to transform that data into interactive charts and filtered views, I was able to surface patterns that would simply not be visible to someone reviewing a spreadsheet or a written report.

The benefits of this approach for road safety policy are considerable. A dashboard like this allows decision makers at the Victorian Department of Transport to immediately see whether crash numbers are trending up or down, how the breakdown of crash severity is changing year by year, which lighting conditions are most associated with crashes, and which road user groups are most frequently involved. All of this can be filtered down to a specific year, severity level, vehicle type, or road user type with a single click. That kind of rapid, evidence-based insight is exactly what is needed for policy decisions about where to direct road safety investment, enforcement resources, and public awareness campaigns.

However, there are real risks associated with data visualisation that I had to think carefully about throughout this project. The most common risk is that a poorly designed visual can mislead the audience, sometimes without anyone intending it to. A bar chart with an axis that does not start at zero, for example, can make a very small change in crash numbers look dramatic and alarming, potentially causing a disproportionate policy response. I tried to address this throughout my dashboard by ensuring all chart axes start at zero, by choosing the most appropriate chart type for each type of data, and by following the design principles outlined by Stephen Few, which emphasise clarity, accuracy, and the avoidance of unnecessary visual decoration.

There are also important limitations in the data itself that need to be acknowledged. Road crash data in Victoria is based on police-attended incidents, which means that minor crashes which are not reported to police are absent from the dataset entirely. This is particularly likely in rural and remote areas where police response is less frequent, which means the dashboard could be understating the true level of crash risk in those regions. There is also an ethical dimension to working with this data. The records in this dataset relate to real individuals who experienced serious and sometimes fatal crashes. Analysing and publishing insights from this data carries a responsibility to treat those individuals with respect and to ensure that the findings are used to improve safety outcomes rather than to make uninformed or sensationalised claims.

Reflecting specifically on my own dashboard, I feel that the combination of the five KPI cards, the annual trend line chart, the severity stacked column chart, the lighting condition bar chart, and the road user comparison chart creates a coherent and useful analytical story about road safety in Victoria. The four slicers for year, severity, road user type, and vehicle type give the user flexible control over the view without the dashboard feeling cluttered. If I were to extend this project further, I would add a geographic map showing crash density across Victorian local government

areas, and a day and time heatmap to identify peak crash windows. Both of these would add substantial value for decisions about enforcement scheduling and targeted infrastructure investment and would bring the dashboard closer to what a professional road safety analyst would produce.